

Experiment 2.2: Design and Performance Evaluation of a Chebyshev Type-I IIR High Pass Filter

MATLAB Code:

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%% Experiment 2.2: Design and Performance Evaluation of a Chebyshev Type-I IIR High Pass Filter

clc;
clear;
close all;

%% Specifications
Fs = 1000;
Fp = 300;
Fs1 = 200;
Rp = 1;
Rs = 40;

%% Normalize
Wp = Fp/(Fs/2);
Ws = Fs1/(Fs/2);

%% Order
[n, Wn] = cheb1ord(Wp, Ws, Rp, Rs);

%% Design
[b, a] = cheby1(n, Rp, Wn, 'high');

%% Responses
[H, f] = freqz(b, a, 1024, Fs);
[h_imp, t] = impz(b, a);
[h_step, t] = stepz(b, a);

%% Plots (3 Left, 3 Right)
figure;

subplot(3,2,1);
plot(f, 20*log10(abs(H)));
grid on;
title('Magnitude Response');

subplot(3,2,2);
plot(f, angle(H));
grid on;
title('Phase Response');

subplot(3,2,3);
stem(t, h_imp, 'filled');
grid on;
title('Impulse Response');

subplot(3,2,4);
stem(t, h_step, 'filled');
grid on;
title('Step Response');

subplot(3,2,5);
zplane(b, a);
title('Pole-Zero Plot');

subplot(3,2,6);
grpdelay(b, a);
title('Group Delay');

disp(['Filter Order = ', num2str(n)]);
```

Output:

